

Evaluation Criteria for Block Cipher Algorithms (Draft)

Institute of Commercial Cryptography Standards

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This document specifies the considerations of evaluating and selecting block cipher algorithms in the Next-generation Commercial Cryptographic Algorithm Program (NGCC) held by the Institute of Commercial Cryptography Standards (ICCS).

1 Security

(1) Overall Structure Security

Algorithms shall have security analyses or proof of the overall structures.

(2) Security Against Known Attacks

The classical security strength of algorithms shall be no less than the bit length of keys against key-recovery attacks. Moreover, the complexity of quantum attacks shall be no less than that of generic ones.

(3) Other Security Properties

Algorithms shall have good statistical randomness and shall be resistant to distinguishing attacks, and etc.

2 Performance

(1) Efficiency

Algorithms will be evaluated based on the computational efficiency of encryption and decryption operations.

(2) Resource Consumption

Algorithms will be evaluated based on the memory cost for software implementations, and the area, time delay, throughput, throughput-area ratio, power consumption, and energy consumption for hardware implementations.

3 Features

(1) Innovativeness

Algorithms are innovative in theory, structure, or other aspects, reflecting new principles and trends in cryptography.

(2) Simplicity

In order to facilitate comprehensive security evaluation, algorithms are easy to understand and implement.

(3) Flexibility

Algorithms can support different key lengths and block lengths easily. Algorithms can be implemented securely and efficiently on a wide variety of platforms, supporting various tradeoffs between efficiency and cost.

(4) Implementation Advantages

Algorithms can support performance advantages on various platforms, including parallel processing, instruction set acceleration, and efficient implementation in resource-constrained environments. Algorithms can have implementation advantages against side-channel attacks and others.

(5) Other Advantages

Algorithms can support the diversity of structure and component selection according to the algorithm design rationale.

4 Overall Considerations

All the aspects including security, performance and features will be taken into account in the evaluation process. Furthermore, commitments made in the intellectual property statements by the submitter(s) and the corresponding patent owner(s) will be considered to be one of the important factors. The algorithms without factors (e.g., intellectual property) that might hinder standardization and widespread adoption will be preferred.